

for tracking. These interfaces have multiple MCUs replaced with a single MCU on a module, so as to reduce the component count and to provide a continuous reading even if the transceiver fails. It also applies to such wearable devices as a wristband to warn against possibility of a heart attack.

Wireless interface module. This module, based on Bluetooth wireless technology, supports display serial interface (DSI) protocol. This decreases the requirement of an external power source, thus further decreasing the number of components on the module. Such modules receive power from software drives, allowing the engineers to design a compact hardware without any requirement to search for an appropriate external power source.

Wireless transceiver with serial data interface. It supports 2x3 multiple input and multiple output (MIMO) configurations for a better array gain and a significant increase in data throughput. This allows the designing of a module without additional frequency requirement and increased transmit power.

Avinash Babu, senior project manager, Mistral Solutions, says, "There has been considerable traction on low-power wireless interfaces like Bluetooth low energy (BTLE or BLE), ZigBee and low-power radio frequency (RF) in the last couple of years. Many of today's low-power wireless solutions are aimed at enabling the Internet of Things."

Wireless microcontrollers

According to Babu, "There are several standard off-the-shelf wireless MCUs available currently, which the designer can use to quickly implement wireless solutions." The recently introduced wireless MCUs based on BLE technology are suitable for systems that have ultra-low-power consumption requirements. These MCUs support different data transfer rates like 250kbps, 500kbps, 1Mbps and 2Mbps, and have good blocking performance and a receiver sensitivity of about -94dBm at 1Mbps. Such MCUs enable long-range applications without any external frontend. These have hardware debug sup-

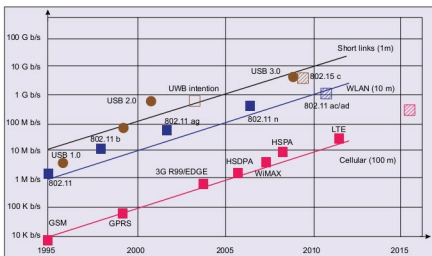


Fig. 2: Trend in wireless communication (Source: Google Search)

port and provide retention to all the relevant registers in all the power modes.

Then there are wireless MCUs suitable for applications related to IoT. This family of MCUs has built-in support for security protocols. Mrinmoy Purkayastha, vice president-Marketing, Calsoft Labs, says, "Security is a very important element in IoT and M2M communications. Devices with embedded hardware accelerators that support different security standards are valuable to build products that are reliable and power-efficient." This family of MCUs is well suited for such applications as heating, ventilation and air-conditioning (HVAC), power-line communication, refrigerators and smart e-meters.

Praveen Ganapathy, director-Processor & Connectivity Applications, TI (India) says, "Design engineers can now design using a single-chip MCU with built-in Wi-Fi connectivity. TI has created a wireless MCU for the Internet of Things, that integrates a high-performance ARM Cortex-M4 MCU, allowing customers to develop an entire application with a single IC. With on-chip Wi-Fi, Internet and robust security protocols, no prior Wi-Fi experience is required. The Wi-Fi network subsystem includes an 802.11 b/g/n radio, baseband and MAC with a powerful crypto engine for fast, secure Internet connections and 256-bit encryption. The Wi-Fi Internet-on-a-chip includes

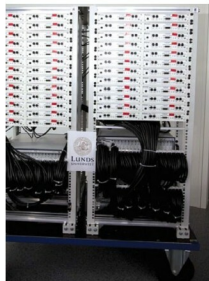


Fig. 3: The massive MIMO testbed at Lund University in Sweden is based on USRP RIO (Source: National Instruments)

embedded transmission control protocol (TCP)/Internet protocol (IP) and transport layer security (TLS)/secure sockets layer (SSL) stacks, hypertext transfer protocol (HTTP) server and multiple Internet protocols."

Smart sensors

The smart and intelligent sensors help reduce the installation and system costs. These increase system flexibility, simplify system deployment and address a new set of applications that was previously impossible with a wired approach. The sensors allow the engineers to design a system with ben-